

3D Probabilistic Gaussian Splatting

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Continuous surfaces

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Soft edges Blended light Depth

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Gaussian splatting models exactly this

Problem with traditional 3D reconstruction:

- Meshes / point clouds = discrete representations
- Leads to:
 - Jagged surfaces
 - Poor continuity
 - Expensive rendering

Goal:

- Continuous, differentiable scene
- Real-time rendering

(Mildenhall et al., 2020; Kerbl et al., 2023)

Key Concept

Represent a scene as a **mixture of 3D Gaussians**

$$p(\mathbf{x}) = \sum_{i=1}^N w_i \mathcal{N}(\mathbf{x} \mid \mu_i, \Sigma_i) \quad (1)$$

- $\mu_i \rightarrow$ position
- $\Sigma_i \rightarrow$ shape + orientation
- $w_i \rightarrow$ importance / opacity

(Kerbl et al., 2023)

Each point becomes a probabilistic blob

- Position: $\mu = (x, y, z)$
- Shape controlled by covariance

$$\Sigma = \begin{bmatrix} \sigma_x^2 & 0 & 0 \\ 0 & \sigma_y^2 & 0 \\ 0 & 0 & \sigma_z^2 \end{bmatrix} \quad (2)$$

- Controls spatial spread
- Enables elliptical (anisotropic) shapes

(Kerbl et al., 2023)

Key Insight

Gaussians remain Gaussians after projection

$$\Sigma' = J\Sigma J^T \quad (3)$$

- J = Jacobian of camera projection
- Result = elliptical splats on screen

(Zwicker et al., 2002; Kerbl et al., 2023)

Each Gaussian contributes to pixel color

$$C = \sum_i T_i \alpha_i c_i \quad (4)$$

- $\alpha_i \rightarrow$ opacity
- $T_i \rightarrow$ visibility
- $c_i \rightarrow$ color

Smooth blending instead of hard surfaces

(Kerbl et al., 2023)

Optimization Objective

Learning Goal

Match rendered image to real image

$$\mathcal{L} = \sum_{\text{pixels}} \|C_{\text{rendered}} - C_{\text{ground truth}}\|^2 \quad (5)$$

We optimize:

- positions μ
- shapes Σ
- colors
- opacities

(Kerbl et al., 2023)

Why It Works

Compared to point clouds

- Smooth instead of noisy
- Continuous surfaces

Compared to NeRF

- Faster
- No heavy ray marching

Real-time: 30–100 FPS with high fidelity

(Kerbl et al., 2023; Mildenhall et al., 2020)

- Dynamic scenes (time-varying Gaussians)
- Integration with AR / VR systems
- Spatial computing with Boxer
 - Real-world + digital scene alignment
 - Scene understanding + interaction layer
- Hybrid models
 - Gaussian splatting + neural fields

(Meta Research, 2024)

References

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